

Joanne Hoar

Technology Educator, Software Developer, Game Developer

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 [xcodegirl](#)

Summary

Canadian Software Developer with 20 years of experience in video game programming, scientific modelling and cross-platform development. Expertise in multiple programming languages, game engines and SDKs. Proven track record in leading teams, mentoring developers, impressing customers and building corporate culture. Strong background in C++, Unreal Engine, iOS and Android using agile methodologies and CI/CD. Passion for algorithmic correctness, mathematics, artificial intelligence and nature.

Experience

Technology Instructor

Jan 2025 – present

Robertson College (contract)

C# / .NET / ASP .NET Core / Java / JavaScript / React / Git / GitHub / Azure / SQL

- Instructed 20+ course offerings across Full Stack Development and Data Analytics diploma programs. Delivered live coding sessions, maintained open-source course materials on GitHub, and achieved an average student satisfaction rating of 4.5/5.0.

Senior Software Developer

2022 – 2025

bitHeads Inc.

Unreal Engine / C++ / Objective-C / Swift / iOS / Android Native / Java / Kotlin / PlayStation / Xbox / Nintendo Switch / Node.js / Groovy / C# / Shell Script

- Developed and maintained cross-platform client SDKs for a backend-as-a-service platform used by 50+ published games across 12 platforms including PlayStation, Xbox, Nintendo Switch, iOS, and Android. Engineered CI/CD and TDD pipelines and managed customer relationships with AAA and Indie studios.

Co-founder/ Lead Developer

2011 – 2022

Silicon Hanna Inc. (self-employed)

Unreal Engine / Objective-C / Swift / iOS / Android Native / Kotlin / Augmented Reality / MacOS / Windows

- Designed, coded and published eleven mobile games for iOS and Android, from concept and game design through to commercial app-store release. Experimented with cutting-edge device technologies including ARKit and other emerging SDKs.

Tools Programmer

2008 – 2011

Fekete Associates Inc. (S&P Global)

C++ / MFC / Windows

- Developed tools that used mathematical modelling to maximize production and improve efficiency for major oil & gas corporations. Led research on a new system that reduced build times by 40%.

Intermediate Software Engineer

2005 – 2008

BJ Pipeline Inspection Services Inc. (Baker Hughes)

C++ / C / OpenGL / PHP / SQL / Neural Networks / Borland / Windows / Linux

- Visualised in-line device data with OpenGL graphics. Implemented neural network algorithms for anomaly detection. Led development of custom web-based GIS reporting software. Improved low-level programming algorithms for data processing.

Co-founder / App Developer / Instructor

2005 – 2008

SuRJE Software Solutions Inc. (self-employed, contract)

Objective-C / C++ / OpenGL / Chipmunk / PHP / JavaScript / Firebase / Google Maps / iOS / Linux

- Created and published physics-based games for iOS. Technical writing, web development, Google Maps and Firebase integration and server administration. Developed course curriculum, instructed C++ programming and assessed student work with the Department of Continuing Education.

Software Researcher / Lead Developer

2005 - 2006

PlayStarMusic Corp.

C++ / PHP / Javascript / .NET / Windows / Linux

- Lead developer and system administrator for an iTunes-like music store. Automated DRM encoding and video streaming. Developed a prototype for a Google-competitive search engine.

Junior Software Engineer

2004 - 2005

Codefast.com

C++ / Qt / Wx / Shell script / Windows / Unix

- Ported applications to operate under a new GUI library. Developed user interfaces for software lifecycle automation.

Web Developer / VR Artist

2002 - 2004

Nickle Arts Museum, University of Calgary (contract)

JavaScript / C++ / OpenGL / PHP / HTML / CSS / 3D Scanning / VR

- Designed and developed an interactive online catalogue for the Gerald Hushlak art exhibit with JavaScript-based features. Contributed original generative art from graduate research. Collaborated on a pioneering VR installation using 3D scanning and CAVE immersion technology.

Research Assistant

2001 - 2004

VLab Computer Graphics Group / Artificial Intelligence Research Hub

C++ / OpenGL / PHP / MySQL / Qt / Solaris / Unix / Linux

- Developed graphics software tools using fractal mathematics and biological modelling. Managed network setup and distributed computing infrastructure. Published five research papers and served as teaching assistant across three courses.

Education

- **Master's Degree** – University of Calgary 2002 – 2004 *Computer Science Thesis: Agent-based Development of Natural Transportation Networks*
- **Bachelor's Degree** – University of Calgary 1998 – 2001 *Computer Science*

Honours & Awards

- Departmental Nomination: Governor General's Gold Medal – Faculty of Graduate Studies 2005
- Postgraduate Scholarship NSERC D3 – Natural Sciences and Engineering Research Council of Canada 2005
- Graduate Student Scholarship – Alberta Learning 2004
- Best Student Paper Award (Co-author) – ACAL 2003
- CPSC Departmental Research Award – University of Calgary 2003

Portfolio

 [xcodegirl/career](https://github.com/xcodegirl/career)

Published Games

Ragin' Bull – iOS, Android (2020)
 AR Picture Portal – iOS (2018)
 Fragmented Reality – iOS (2018)
 IceCream AR – iOS (2018)
 Jewel Slide – iOS, Android (2017, 2019)
 Word Slide & Find – iOS (2017)
 A Word Theme – iOS, Android (2017, 2019)
 CryptoQuip – iOS, Android (2011, 2019)

Rafter's Loot – iOS (2011)
Star Chase – iOS (2008)
Bitsy – iOS (2008)